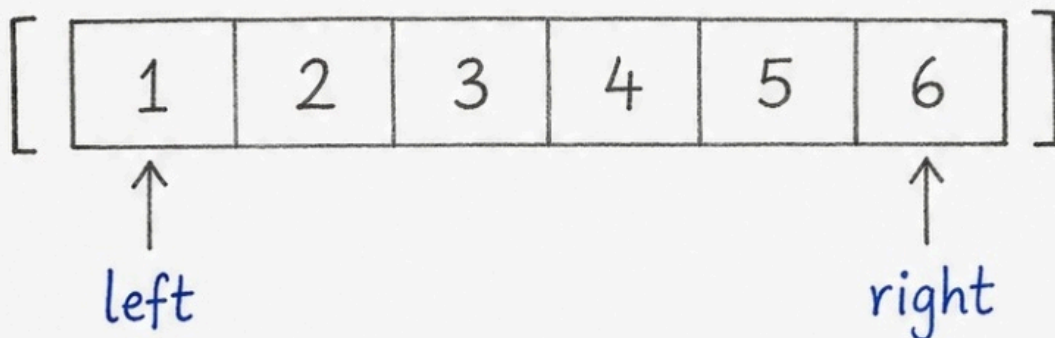


TWO POINTERS TECHNIQUE

Concept, Types, Examples & Interview Questions

A powerful technique to solve array, string and linked list problems efficiently ✧



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Usually helps reduce time complexity
from $O(n^2)$ to $O(n)$

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WHAT IS TWO POINTERS?

Two pointers is a problem-solving technique where two variables traverse the same array, string, or linked list.

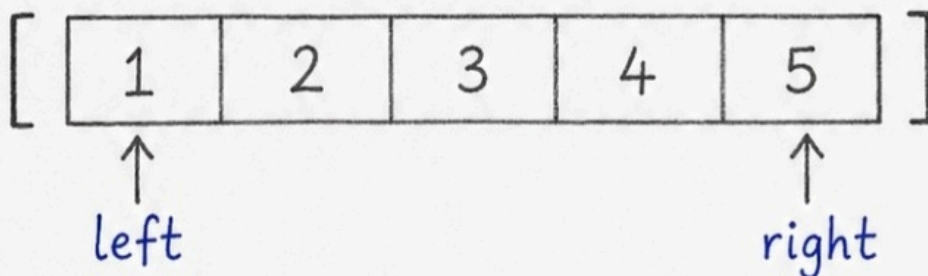
The pointers may start from opposite ends or may move in the same direction.

Common pointer names:

1. left - right
2. start - end
3. slow - fast
4. i - j

Main goal:

Solve problems
efficiently with less
extra space



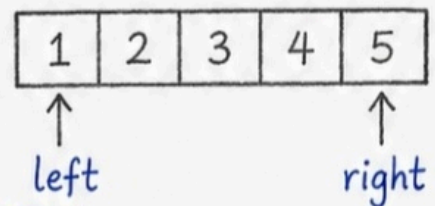
Move the pointers based on a condition
until the answer is found.

WHEN TO USE TWO POINTERS?

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Use this technique in problems like:

- Sorted array + target sum
- Palindrome checking
- Reversing array or string
- Remove duplicates from sorted array
- Move zeroes to the end
- Linked list cycle detection
- Finding middle of linked list
- Sliding window / longest substring problems



Why it is useful:

- ✧ Reduces unnecessary comparisons
- ✧ Often gives $O(n)$ time
- ✧ Usually uses $O(1)$ extra space

Interview clue: If the question asks for $O(n)$ time and $O(1)$ space, think about two pointers.



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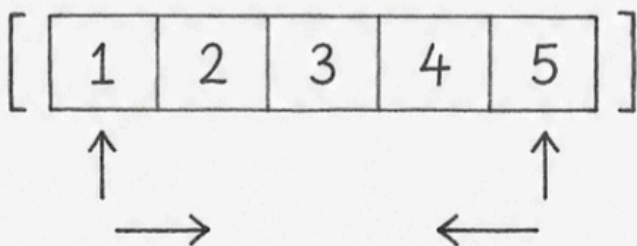


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TYPES OF TWO POINTERS

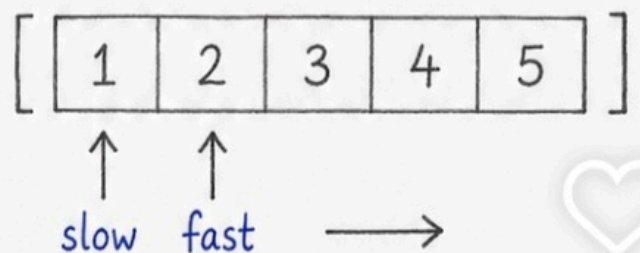
1) Opposite Direction

- One pointer starts from the beginning
- Other pointer starts from the end
- Both move towards each other
- Used in: palindrome, reverse array, pair sum in sorted array



2) Same Direction

- Both pointers move forward
- Often called slow and fast
- Used in: move zeroes, remove duplicates, linked list cycle



Choose pointer movement according to the problem condition.

TIME COMPLEXITY + ADVANTAGES

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1. Complexity

- Most two pointer problems run in $O(n)$ time
- Extra space is usually $O(1)$
- Reason: the data is often scanned only once

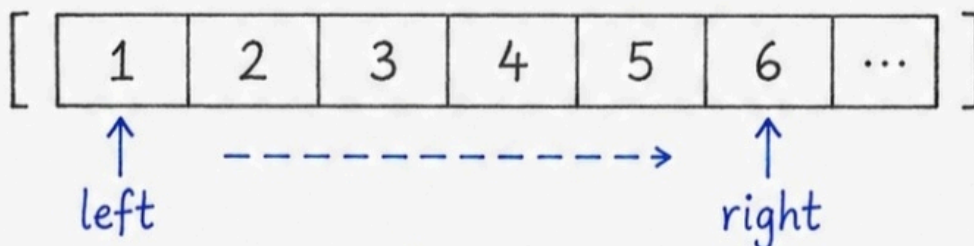
2. Advantages

- Faster than brute force approach
- Easy to understand and implement
- Good for in-place operations
- Saves extra memory
- Very common in coding interviews
- Useful for arrays, strings and linked lists

⇒

Two pointers often converts an $O(n^2)$ idea into an $O(n)$ solution.

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